

RESEARCH ARTICLE

Applying the N-mixture model approach to estimate mosquito population absolute abundance from monitoring data

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Abstract

1. Estimating population abundance is a key objective of surveillance programmes, particularly for vector species of public health interest. For mosquitos, which are vectors of human pathogens, established methods to measure absolute population abundance such as mark-release-recapture are difficult to implement and usually spatially limited. Typically, regional monitoring schemes assess species relative abundance (counting captured individuals) to prioritize control efforts and study species distribution. However, assessing absolute abundance is crucial when the focus is on pathogen transmission by contacts between vectors and hosts. Here, we applied the N-mixture model approach to estimate mosquito abundance from standard monitoring data.
2. We extended the N-mixture model approach in a Bayesian framework by considering a beta-binomial distribution for the detection process. We ran a simulation study to explore model performance under a low detection probability, a time-varying population and different sets of independent variables.
3. When informative priors were used and the model was well specified, estimates by N-mixture model well correlated (>0.9) with synthetic data and had a mean absolute deviation of about 20%. Correlation decreased and biased increased with uninformative priors or model misspecification.
4. When fed with field monitoring data to estimate the absolute abundance of the mosquito arbovirus vector *Aedes albopictus* within the metropolitan city of Rome (Italy), the N-mixture model showed higher population size in residential neighbourhoods than in large green areas and revealed that traps located adjacent to vegetated sites have a higher probability of capturing mosquitoes.
5. *Synthesis and applications.* Our results show that, if supported by a good knowledge of the target species biology and by informative priors (e.g. from previous studies of capture rates), the N-mixture model represents a valuable tool to exploit field monitoring data to estimate absolute abundance of disease vectors and to assess vector-related health risk on a wide spatial and temporal scale. For mosquitoes specifically, it is also valuable to invest in increased efficiency of trapping devices to improve estimates of absolute abundance from the models.

KEYWORDS

abundance, Bayesian model, disease vector, N-mixture model, tiger mosquito, trap efficiency, vector-borne pathogens

1 | INTRODUCTION

Estimating the absolute abundance of animal populations is an extremely challenging task. A trade-off exists between sampling efforts, analytical tools and information gathered. On the other hand, estimates of relative abundance or indexes of abundance are more easily achievable, especially when dealing with arthropod populations. These approaches rely on the assumption that “the higher the number of collected specimens, the higher the population abundance” and does not take into account differences in detectability and/or trap performance, de facto providing biased estimates when factors influencing the detection are not controlled for (Joseph, Elkin, Martin, & Possingham, 2009).

In the case of insect species of public health relevance, such mosquito vectors of malaria parasites or of arboviruses, absolute abundance is a crucial parameter to infer the vector to host contact ratio and inform models aimed at estimating risk of pathogen transmission and effectiveness of vector control interventions. However, Mark-Release-Recapture experiments – which are considered the gold standard to estimate size of animal populations – are not only very laborious and challenging to implement in the field, but also raise ethical concerns due to the need to release large number of potential vectors, which may contribute to disease transmission. For this reason, only few studies have been carried out so far (Cianci et al., 2013; Gouagna, Dehecq, Fontenille, Dumont, & Boyer, 2015; Villela et al., 2017, 2015) to estimate the absolute abundance of mosquito vectors which cause millions of infections and thousands of deaths every year (WHO, 2016).

The N-mixture model approach is a statistical method exploited in ecological studies to estimate absolute population abundance from observed field data (Kéry & Royle, 2016; Royle & Dorazio, 2009). This method treats data as the observable outcome of two linked components: an observation and a population process. Therefore, this approach takes simultaneously into account the underlying ecological process and the mechanism by which observations are sampled. Nowadays, it has been extended to deal with: non-independent detection that may occur when individuals show correlated behaviour (Martin et al., 2011), spatio-temporal variation (Hostetler & Chandler, 2015); species uncertainty that may occur when it is difficult to exactly identify captured individuals (Chambert, Hossack, Fishback, & Davenport, 2016); and density dependence and environmental stochasticity (Bellier, Kéry, & Schaub, 2016). In recent years, it has been applied to various animal species (Belant et al., 2016; Hunter, Nibbelink, & Cooper, 2017; Kéry et al., 2009), but it is still considered an emerging method (Dénes, Silveira, & Beissinger, 2015) and its reliability and robustness have been questioned (Barker, Schofield, Link, & Sauer, 2017; Link, Schofield, Barker, & Sauer, 2018).

We here proposed and tested the N-mixture model to estimate absolute mosquito abundance exploiting monitoring data routinely gathered in surveillance schemes carried out in areas where mosquitoes represent either a public health risk or a severe nuisance.

Firstly, we tested how well the N-mixture approach performs to estimate a set of parameters and population absolute abundances from synthetic trap data under realistic scenarios of environmental conditions. We addressed how robust the approach is against the typical constraints of a mosquito field monitoring scheme (i.e. low detection/capture rate, repeated multiple sampling dates and violation of the assumption of closed population) and developed a comprehensive simulation study to assess the robustness of the approach against (a) the introduction of additional unexplained variation, and (b) under-parametrization of the model by not considering relevant covariates in the observation or in the population process.

Secondly, we applied the N-mixture model approach to a case study to investigate whether the abundance of the tiger mosquito *Aedes albopictus* differs in different ecological contexts within the city of Rome (Italy) and to explore the implications for vector control. In recent decades, this aggressive day-time biting species has become a global public health threat due to its invasive potential which extended worldwide from Asia and to its capacity to transmit a large number of arbovirus (such as Chikungunya, Dengue and Zika; Gratz, 2004). The species has been well established in Italy for more than 20 years and has been responsible for two Chikungunya outbreaks causing 250 and 500 cases (Istituto Superiore di Sanità, 2017; Manica et al., 2017; Rezza et al., 2007).

2 | MATERIALS AND METHODS

2.1 | N-mixture model approach

The model we proposed is framed in a Bayesian hierarchical perspective, which allows extreme flexibility and could accommodate further extension of the N-mixture model including spatial dependency or nonlinear effects.

We consider specimen counts n_{ijt} collected in the trap i of site j at time of collection t , sampled from a Binomial distribution $n_{ijt} \sim \text{Binomial}(\pi_{ijt}, N_{jt})$, where:

- (i) π_{ijt} is the capture/detection probability of trap i within site j at time of collection t that is assumed to come from a Beta distribution $\pi_{ijt} \sim \text{Beta}(a_{ijt}, b_{ijt})$;
- (ii) N_{jt} is the unobserved population absolute abundance present within site j at time of collection t , that is assumed to be a discrete number sampled from a Poisson distribution with mean λ_{jt} , i.e. $N_{jt} \sim \text{Poisson}(\lambda_{jt})$.

Therefore, we assume that repeated observations from the same site j and time t are drawn from the same subpopulation N_{jt} . The mean of the Beta distribution represents the average capture/detection probability and is indicated as P_{ijt} . The means of the Poisson and Beta distribution could be modelled as a function of covariates.

Specifically, for the Poisson distribution we assume that $\log(\lambda_{ijt}) = \mathbf{Y}\boldsymbol{\beta} + \varepsilon_j$, where $\mathbf{Y}$ is a matrix whose elements are the covariate values, $\boldsymbol{\beta}$ is the vector of estimated parameter and $\varepsilon_j \sim \text{Norm}(0, \sigma^2)$ are site-dependent residuals.

For the Beta distribution, we use the following parametrization: $a_{ijt} = \vartheta P_{ijt}$ and $b_{ijt} = \vartheta(1 - P_{ijt})$. Under this parametrization the mean of the Beta distribution is equal to P_{ijt} . Then it is easy to specify a linear model on P_{ijt} through the logit transformation so that $\text{logit}(P_{ijt}) = \mathbf{X}\boldsymbol{\gamma}$ where $\text{logit}(P_{ijt}) = \log(P_{ijt}/(1 - P_{ijt}))$, $\mathbf{X}$ is the covariates matrix, which can impact trap performances and therefore the detection process, and $\boldsymbol{\gamma}$ is the vector of estimated parameter.

A Multivariate Normal distribution ($\text{MVN}(\mathbf{0}, 1000 \times \mathbf{I})$) was chosen as minimally informative prior for the parameters of the population process ($\boldsymbol{\beta}$) and for the covariates of the detection process.

On the other hand, informative priors were chosen for the intercepts of the detection processes ($\gamma_0 \sim \text{Norm}(-7, 0.1)$) as well as for the parameter ϑ of the Beta distribution: $\vartheta \sim \text{Gamma}(60, 0.1)$. Parameters values for the informative priors were based upon estimates obtained in a Mark-Release-Recapture experiment on *Aedes albopictus* in Rome (Marini, Caputo, Pombi, Tarsitani, & Della Torre, 2010) where the same traps (ST) were used (see Appendix S1).

Finally a Half-Cauchy distribution σ was considered ($\sigma \sim |\text{Norm}(0, 1)|/\text{Norm}(0, 25)$) for the standard deviation of the Normal distribution used to account for unexplained variation among sites (i.e. random effects). Bayesian inference was carried out using Markov Chain Monte Carlo (MCMC) simulative approach using R version 3.4.0 (R Core Team, 2017) and Jags version 4.2.0 (Plummer, 2003; Su & Yajima, 2015).

2.2 | Creating synthetic data

We generated synthetic data for the covariates and applied the previous model to obtain synthetic data of trap mosquito collections.

At first, we considered three covariates (Table 1) for the population abundance process ($\mathbf{Y}$) and two for the detection/capture process ($\mathbf{X}$) considering five traps ($i = 1, \dots, 5$) for each of the 12 sites ($j = 1, \dots, 12$) at 12 weeks of collections ($t = 1, \dots, 12$).

In order to simulate a mosquito population temporal profile, realistic temperature and precipitation data (see section *Applying N-mixture model to observed data* for details) were used as a basis for creating synthetic data for the following two covariates of the population process:

- weekly mean Land Surface Temperature (LST) recorded from 15 July 2013 to 30 September 2013 in Rome to which we added a site dependent and a collection dependent noise drawn from two Normal distributions of mean 0 and standard deviation 2 and 1, respectively;
- cumulative precipitation in the previous 4 weeks of collection recorded from 15 July 2013 to 30 September 2013 in Rome as suggested in Manica et al. (2016).

The third covariate was sampled from a Uniform distribution (min = 0.2, max = 0.8) and kept constant over collection to represent a physical characteristic of the site.

Similarly, realistic data related to residual water level in trap was used to create synthetic data for one of the covariates related to the detection process. The weekly mean residual water level in trap (measured in millilitre, see section *Applying N-mixture model to observed data* for details) recorded from 15 July 2013 to 30 September 2013 in Rome, was used, adding a trap- and a collection-dependent noise drawn from two Normal distributions: $\text{Norm}(0, 20)$ and $\text{Norm}(0, 50)$ respectively.

The other covariate of the detection process was sampled from a Uniform distribution (min = 0, max = 1) and kept constant over collections to characterize the ecological features of the trapping area.

The parameters values for the population process ($\boldsymbol{\beta}$) were chosen as follows: $\beta_0 = 0$, $\beta_1 = 1.5$, $\beta_2 = 0.75$, $\beta_3 = -0.75$. Values for the site-dependent residuals ε_j were sampled from a Normal distribution $\text{Norm}(0, \sigma = 0.5)$.

TABLE 1 Model variables and parameters used in the simulation of field monitoring of mosquito abundance

Process	Model parameter	Parameter value	Variable	Description of synthetic data
Population	β_0	0		Intercept of population process
	β_1	1.5	Y_1	Derived from land surface temperature data
	β_2	0.75	Y_2	Derived from precipitation data
	β_3	-0.75	Y_3	Sampled from Uniform distribution
	σ	0.5		Standard deviation for site-dependent noise
Detection	γ_0	-6.5		Intercept of detection process
	γ_1	0.5	X_1	Derived from residual water level in trap
	γ_2	-0.5	X_2	Sampled from Uniform distribution
	ϑ	500		Gamma distribution parameter

Parameter values for the detection process (γ) were chosen as follows: $\gamma_0 = -6.5$, $\gamma_1 = 0.5$, $\gamma_2 = -0.5$, while ϑ was set at 500.

All variables were standardized (subtracted their mean and divided by their standard deviation) to help mixing of chains and interpretation of results (Schielzeth, 2010) with the exception of Y_1 (weekly mean LST), which was centred around 13°C and then divided by its standard deviation to impose a biological constrain on the effect of temperature on population abundance, as 13°C is the lower temperature threshold for emerging *Ae. albopictus* females (Roiz, Rosà, Arnoldi, & Rizzoli, 2010). Therefore, we fitted the model for synthetic data without intercept, forcing the regression line to go through the origin at 13°C in adherence to *Ae. albopictus* mosquito biology.

Given the defined parameter and covariate values for the population process, we obtained the population absolute abundance present within site j at time of collection t (N_{jt}) by sampling from a Poisson distribution (see section *N-mixture model approach*).

Specimen counts data (n_{ijt}) were obtained sampling from the Binomial distribution, describing the detection process. Then, to account for random variation in the detection process, the specimen counts sampling was repeated 100 times, thus obtaining 100 different datasets mimicking different monitoring outcomes from the same population.

2.3 | Simulation study on synthetic data

A set of simulations was carried out in order to test the robustness of the chosen N-mixture model approach under the following assumption: extremely low detection/capture rate and time-varying population abundance.

We fitted the N-mixture model to 100 synthetic datasets in order to estimate population abundance (N) along with all parameters for both the detection and the population process. Precisely, we computed for each parameter: the distribution of the mean of the 100 posterior distributions, the coverage, i.e. the proportion of posterior distributions where 95% credible interval contains the synthetic parameter values (see Table 1) and the root mean squared error (RMSE). For each of the 100 simulation runs, three parallel MCMC chains were run, each with 40,000 iterations. The first 10,000 were discarded as burn-in. A thinning rate of one on 10 was applied resulting in 9,000 iterations.

In addition, we investigated how the N-mixture model would perform under different model specifications (Knape & Korner-Nievergelt, 2015; Link et al., 2018):

- (i) when the population process is under-parametrized, so not all the covariates used to simulate population abundance are included in the model (i.e. discarding the second and third covariate in turn),
- (ii) when the detection process is under-parametrized, so not all the covariates used to simulate specimen counts are included in the model (i.e. discarding one covariate in turn),
- (iii) when additional variation sampled from Norm(0,0.1) is used to simulate specimen counts but is not accounted for in the N-mixture model used to estimate parameters and population abundances,

- (iv) when non-informative priors are chosen for the γ and ϑ parameter of the detection process.

2.4 | Applying N-mixture model to observed data

Mosquito collections were carried out weekly from 1 July 2013 to 20 November 2013 ($t = 1, \dots, 22$ weeks). Twelve sites ($j = 1, \dots, 12$) were identified inside the metropolitan area of Rome (Italy) and clustered in four zones (named S1, S2, S3, S4). Three sites were selected within each zone. Each site encloses a circular area of 300 m radius likely representing three different ecological habitats for *Aedes albopictus*.

We arbitrarily defined the three habitats as:

1. "Vegetated" characterized by urban green spaces with high presence of grasslands (>50%) and trees (from 30% to 40%), few buildings or roads and scarcely inhabited;
2. "Mixed" characterized by presence of buildings (from 19% to 27%) and trees (from 23% to 30%);
3. "Residential" with similar characteristics to "Mixed" but more densely populated (Figure S1).

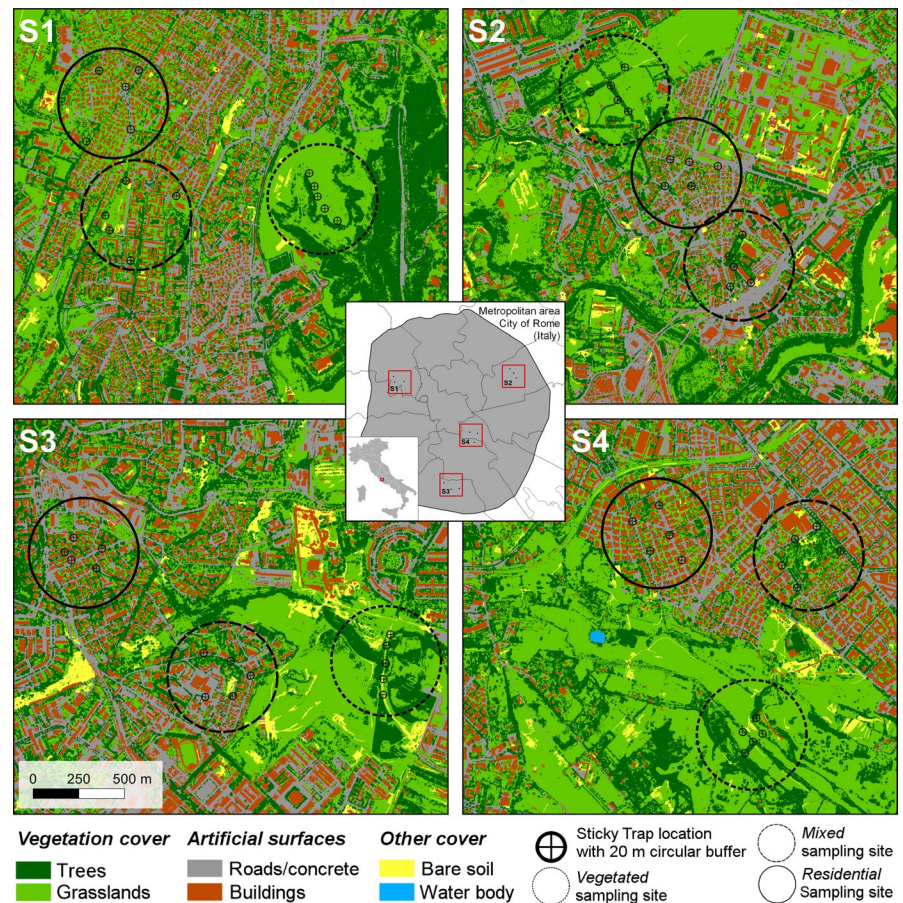
Ecological habitats were identified by visual inspection of a qualified medical entomologist (author BC) and by quantitative assessment carried out both from population census data (ISTAT, 2011) and from a series of spatial datasets. Descriptive meteorological and land cover variables were recorded both at site level (300 m circular buffer) and at trap level (20 m circular buffer), using the methodology described in Manica et al. (2016). Land Surface Temperature (LST) recorded at each site has been extracted from reconstructed temporal series of MODIS satellite data, collected by NASA (<http://modis.gsfc.nasa.gov>) and processed as described in Metz, Rocchini, and Neteler (2014). The cumulated mm of precipitation at each trap location was derived from the spatial interpolation of mm of daily precipitation data recorded at 46 meteorological sampling stations, collected by the Hydrographic Service of Lazio Region and disseminated through the hydrographic annals (<http://www.idrografico.regione.lazio.it/annali/index.htm>).

Sticky Traps (ST) (Facchinelli et al., 2007) were located in the four zones (S1–S4) and within the three different habitats (Vegetated, Mixed, Residential) as shown in Figure 1. Five STs ($i = 1, \dots, 5$) were located within each site (15 traps per zone) and equipped with 500 ml of tap water and sticky sheets. Every week, the residual trap water level (ml) was recorded and mosquito collected, screened for species and gender and counted. Afterwards, STs were replenished with 500 ml of tap water and equipped with new sticky sheets.

The weekly number of captured adult *Ae. albopictus* females in each ST (Figure 2) was used as response variable in the model.

The covariates we considered for the capture/detection process were percent of trees within the 20 m circular buffer and the weekly residual water in STs (mm). For the population abundance process we considered habitat type and climatic covariates (e.g. temperature and precipitation) preceding the sampling week, to take into account the effect of climate on the development of mosquito immature

FIGURE 1 Location of the four sampling zones in the metropolitan area of the city of Rome (red square in central panel) and the Sticky Traps (cross circles) for the 12 sampling sites (circles) over the different land cover classes



stages (Roiz et al., 2010). Specifically, the climatic covariates were the mean LST recorded within the time window including the 3rd and 4th weeks prior to the collection, its quadratic term (to account for a nonlinear relationship) and the cumulated precipitation (mm) in the previous two weeks. Within two “Vegetated” sites for two weeks (34th and 35th), all STs were found deactivated, meaning that both mosquito data and residual water level were missing (Figure 2). To overcome this issue, for those weeks and sites we simulated a single ST assigning the average covariate values recorded on active STs in the same weeks. Finally, we assumed that the capture rate of the traps is an average value over a monitored area defined by mosquito flight range. Then we used the estimate of mosquito absolute abundance produced by the N-mixture model to compute mosquito density (per hectare) over the monitored area.

All quantitative variables were standardized (subtracted their mean and divided by their standard deviation) to help chains mixing and ease interpretation of results (Schielzeth, 2010). The model featured a burn-in of 250,000 iterations, a thinning rate of 150, three chains and a total number of 500,000 iterations, resulting in 5,001 iterations per parameter for each posterior distribution. From the estimated mosquito absolute abundance, we computed the vector to host ratio, dividing mosquito abundance by the resident human population (ISTAT, 2011) in a 300-metre buffer.

Assessment of mixing of chains and model statistical assumptions was carried out by graphical analysis of residuals and simulations

from model estimated parameters (Zuur, Ieno, & Freckleton, 2016). Finally, to further assess model performance we compared model posterior predictive distribution to observed data.

3 | RESULTS

3.1 | Simulation study on synthetic data

Fitting the N-mixture model to synthetic data we showed that all parameters for both the detection and the population process are within the 95% CI of the distribution of the mean estimated values except for γ_2 (Table 2). All the parameters for the population abundance process had >98% coverage while coverage was lower for the detection process, especially for γ_2 (71%).

The synthetic population data are consistently within the 95% credible interval of the estimated population, leading to a coverage above 90% for each site, although for a few sites (Site-2, Site-10, Site-11) the estimated population abundance is slightly higher than the created synthetic value (Figure 3).

Regarding the covariates for the population absolute abundance, failing to include Y_3 , a covariate representing a physical characteristic of the site that was kept constant over time, did not affect greatly the results. However, not including the spatio-temporal covariate in the population process (precipitation) leads to a worse representation of the population dynamics and to more biased estimates of absolute

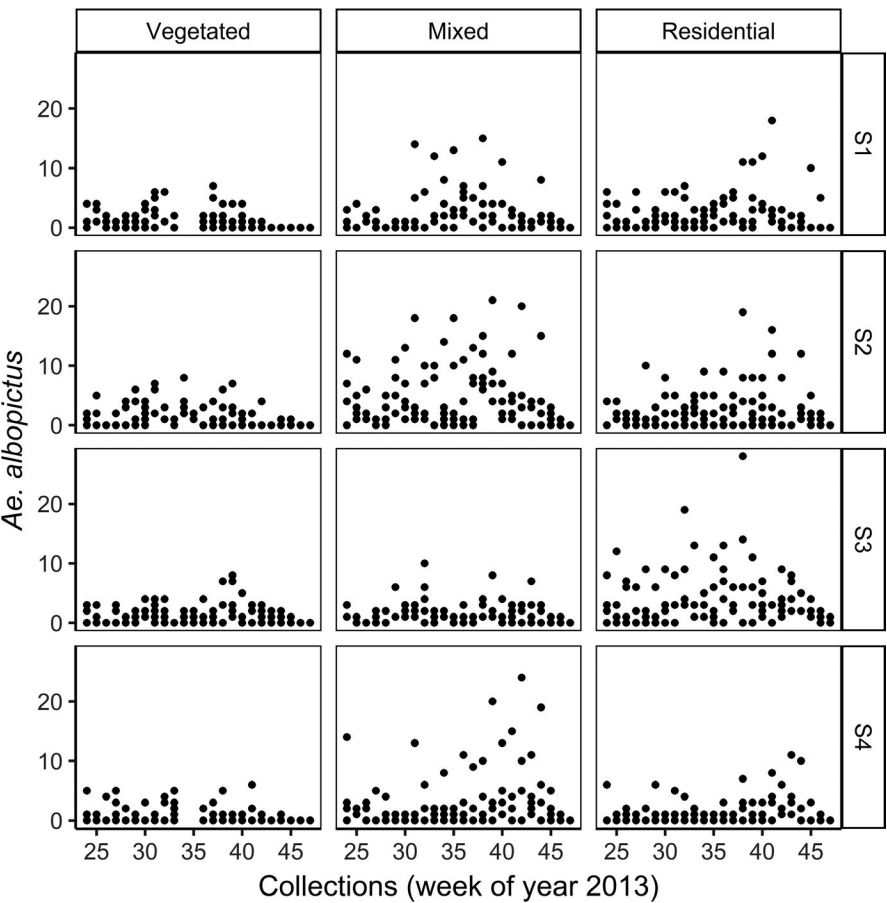


FIGURE 2 Weekly trap collection (black dots), in each sampling zone (panel S1–S4) for each habitat (Residential, Mixed and Vegetated) in Rome, Italy. Each panel represents one of the twelve sites considered in the analysis. The x-axis shows the week of collection, while the y-axis shows the counts of *Aedes albopictus* females trapped in each of five Sticky Traps

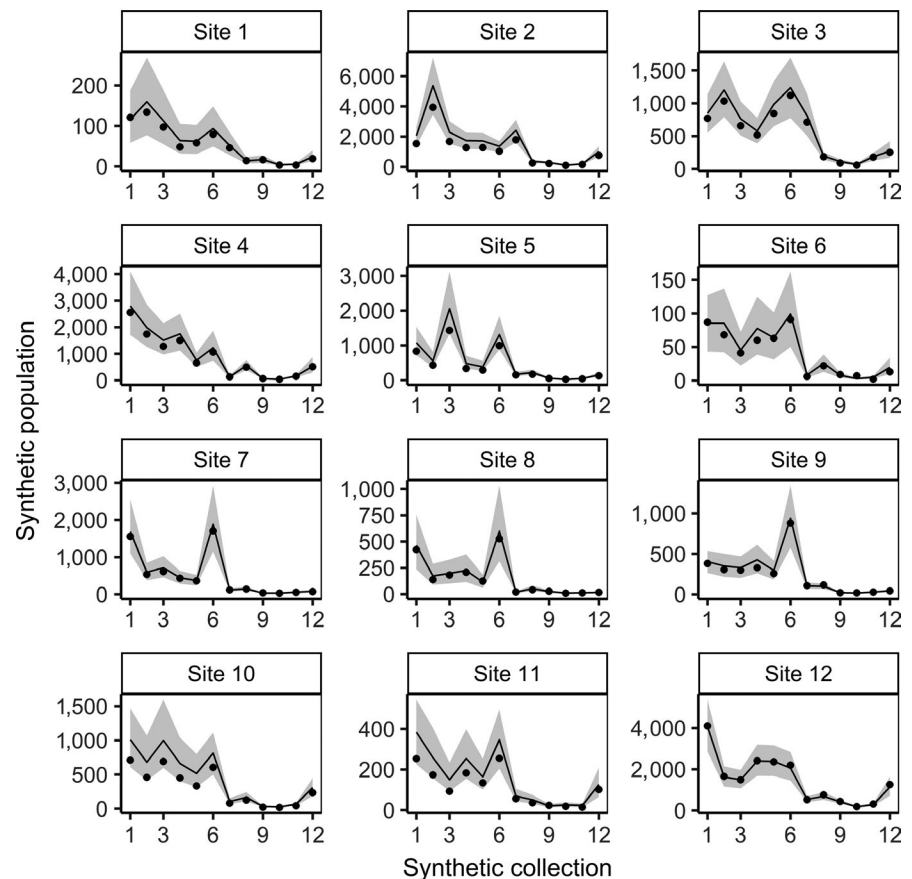
TABLE 2 Result of simulation of the model including all covariates and no additional noise

Process	Parameter	Synthetic value	Estimated value from simulations			Coverage (%)	RMSE
			Mean	0.025% CI	0.975% CI		
Population	β_1	1.50	1.517	1.442	1.591	99	0.035
	β_2	0.75	0.734	0.612	0.851	98	0.051
	β_3	−0.75	−0.618	−0.796	−0.449	100	0.140
	σ	0.50	0.483	0.304	0.678	100	0.076
Detection	γ_0	−6.50	−6.698	−6.923	−6.424	90	0.211
	γ_1	0.50	0.598	0.496	0.722	91	0.100
	γ_2	−0.50	−0.625	−0.721	−0.523	71	0.126
	ϑ	500	596.55	565.49	632.10	99	96.55

abundance that for some site were consistently outside the 95% credible interval; for some sites this happens as frequently as 99% of outputs (Appendix S2, simulation II). Fitting the same model considering also the intercept in the model specification (with vague prior) leads to comparable results. Moreover, even if the model computed reasonable estimates of the population absolute abundance, excluding the spatio-temporal variables in the detection/capture process led to extremely biased model parameters (“true” value never within the credible interval in the 100 simulation runs). A similar but less strong pattern could be observed when neglecting the spatio-temporal covariate in the capture/detection process (water). Finally,

failing to include the covariate in the capture/detection process that was kept constant over time had a negative effect only in few sites without affecting the estimation of model parameters (Appendix S2, simulation IV, V). In our simulation study, the main consequences of adding unaccounted random noise in the detection/capture process were a reduction in the accuracy of the estimation of γ_0 , the overestimation of the mosquito population on average and an increase of the RMSE in some sites (Appendix S2, simulation VI–X). This behaviour is consistent with that observed elsewhere (Link et al., 2018) and may constitute one of the major drawbacks of the N-mixture model approach given that misspecification and unaccounted variance

FIGURE 3 Population absolute abundance comparison: synthetic and fitted by the N-mixture model. Dots represent the synthetic population absolute abundance, the solid line represents the posterior mean population absolute abundance estimated from the model and averaged over the 100 runs, grey areas represent the 95% credible interval. The x-axis shows the synthetic time of collections, while the y-axis shows the absolute population abundance; note that each panel has a different y-axis range to ease visualization



are likely to occur in real-case studies. However, the proportion of times the synthetic values of the population absolute abundance fell within the 95% was not considerably affected, at least for the simulated intensity of the noise.

3.2 | N-mixture model results on observed data

In the case study a total of 2,504 *Aedes albopictus* females were collected, the N-mixture model showed a quadratic relationship of the mosquito population abundance with LST, with a peak at about 25°C (Table 3). On the other hand, the cumulative precipitation in the two weeks preceding the collections showed a negative association with population absolute abundance. Vegetated habitat had statistically lower infestation than the other two habitats, while the effect of Mixed and Residential habitat on *Ae. albopictus* population absolute abundance was comparable.

Regarding the detection process, water evaporation in ST (as assessed by weekly measures of residual water) was negatively associated to mosquito counts, while the higher abundance of vegetated habitats had a positive effect size. The low value of the intercept for the detection/capture process indicates a very low capture rate of STs (Table 3). The estimated capture rate on average was 0.022% (95% ci: 0.017%–0.026%) which is lower than previously observed in a Mark-Release-Recapture study (mean capture rates of three releases: 0.082%, 0.093% and 0.059%, number of traps = 55, Figure S2).

Estimates of population absolute abundance provided mean values ranging from 9.6 to 816.9 female mosquitoes/hectare overall, and from 22 to 800, 20.8 to 816.9, and 9.6 to 251 in residential, mixed and vegetated habitats respectively (Figure 4).

Model validation indicated that the proposed model complied with underlying statistical assumptions and simulated data from the model reasonably resembled observed data (Figures S3–S5).

4 | DISCUSSION

Results show that the N-mixture model is a promising framework to estimate absolute mosquito abundance based on data from routinely monitoring activities, so far only exploited to estimate relative abundance. This could lead to a novel way to obtain more realistic estimates of vector-host ratio needed to predict the risk of pathogen transmission, as well as of effectiveness of control interventions, at a local level.

Following the recommendation of Barker et al. (2017) and Kéry (2018), the model incorporated informative priors on capture rate (Marini et al., 2010) and mosquito biology (Roiz et al., 2010) to overcome the inherent biases linked to the N-mixture approach as discussed in Link et al. (2018) and Duarte, Adams, and Peterson (2018). It should be noted that choice of priors does have an impact on the final estimates and should be informed by evidence. In particular, the intercept of the detection/capture process may be

TABLE 3 N-mixture model result on the *Aedes albopictus* case study. LST = Land Temperature Surface, Vegetated, Mixed and Residential refer to Habitat Type

Process	Covariate	Mean	SE	2.5%	97.5%
Population	Mixed (intercept)	9.428	0.26	8.91	9.939
	LST	2.523	0.646	1.238	3.74
	LST ²	-2.251	0.636	-3.46	-0.982
	Precipitation	-0.375	0.055	-0.476	-0.27
	Vegetated	-0.93	0.343	-1.636	-0.241
	Residential	0.054	0.348	-0.659	0.746
	Difference between Residential and Vegetated	0.984	0.349	0.289	1.700
	σ	0.441	0.147	0.248	0.808
Detection	Intercept	-8.447	0.112	-8.659	-8.255
	Residual Water	0.119	0.04	0.04	0.198
	Tree (%)	0.118	0.04	0.035	0.197
	ϑ	4,360.55	489.13	3,516.56	5,381.98

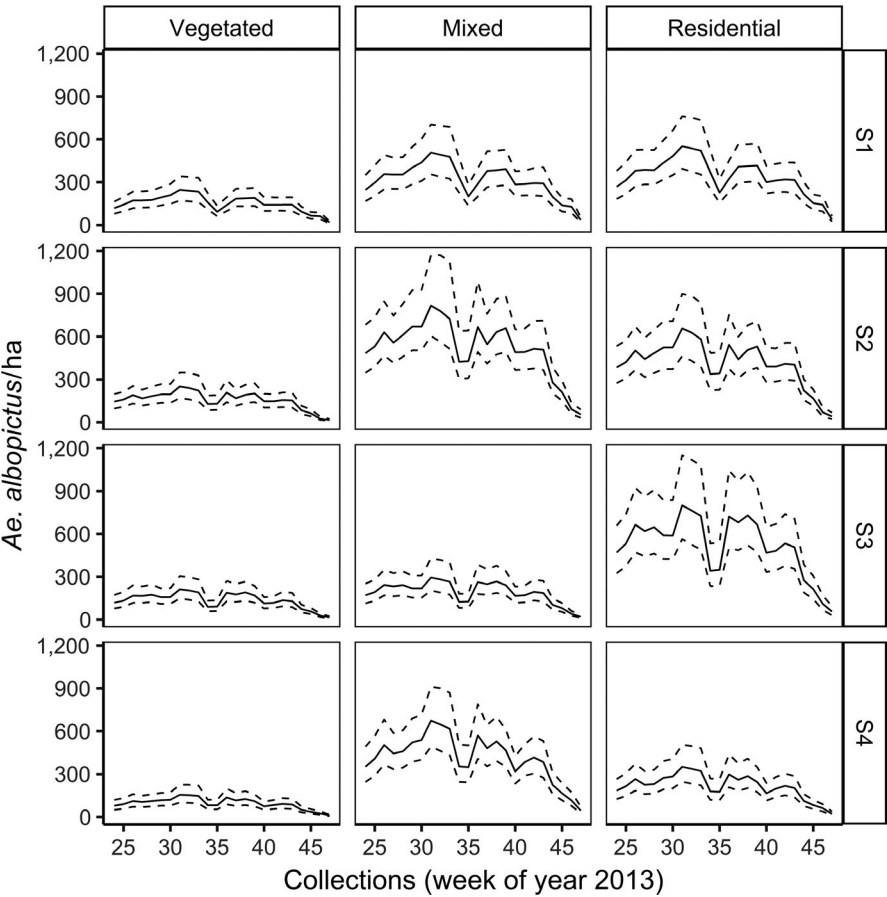


FIGURE 4 Case study results of *Aedes albopictus* population density estimated by the N-mixture model in the twelve sites (300 m radius buffer), in each sampling zone (panel S1–S4) for each habitat (Residential, Mixed, Vegetated) in Rome, Italy. Each panel represents one of the twelve sites considered in the analysis. The x-axis shows the week of collection, while the y-axis shows the estimated density per hectare of *Aedes albopictus* females. The solid lines represent the mean value of the posterior distribution of the absolute abundance divided per the site area, the dashed lines represent the 95% credible interval

difficult to estimate as fluctuating together with other covariates; despite assuming a very informative prior may help identifiability, nevertheless it may also force a specific result. Our results show that running the model with these informative priors produced less biased estimates of mosquito population absolute abundance with respect to using non-informative priors and that neglecting

some important covariates could lead to biased estimates (see Appendix S2, section Simulation Result). Therefore, although in principle the N-mixture approach could be applicable in any areas based on routine mosquito monitoring data, the need of clear assumptions on prior distributions and model specification (Knappe & Korner-Nievergelt, 2015) reduce its application in the real-world.

Additionally, separating the effect of variables between the population and the detection/capture process could represent an additional challenge.

We are aware that the N-mixture model is available in a frequentist framework through software packages like *unmarked* (Fiske & Chandler, 2011). However, this specification may lead to unrealistic (large) estimate of detection probability as the model tends to miss estimates of detection rate near the boundaries (0 or 1), which results in biases when it is known that the detection rate is very low. We were able to overcome the issue by using a Bayesian perspective, tailoring the prior distribution of the detection/capture rate parameter on previous results obtained in Mark-Release-Recapture experiments and taking into account mosquito biology to specify the model. This approach could be extended to other settings, mosquito vector species or trap devices, but efforts should be made to correctly specify the model, to assess trap performances and its priors distribution given the emerging evidences that N-mixture model misspecification will result in low accuracy and poor reliability compared to standard regression (Barker et al., 2017; Link et al., 2018). The N-mixture model has been very recently implemented in the Integrated Nested Laplace Approximations (Rue et al., 2017) which might reduce computation time (Meehan, Michel, & Rue, 2017) and make it more appealing to a broader audience.

The analysis of our case study showed that absolute *Ae. albopictus* abundance was higher in densely populated areas (i.e. Residential and Mixed Habitats) than in highly vegetated and less populated ones. Our estimate of *Ae. albopictus* densities is relatively lower than those estimated by Manica et al. (2017) in the same region. However, a crude comparison may not be appropriate due to spatial and temporal heterogeneities arising from different sampling locations. Whenever feasible, it would be beneficial to set up a Mark-Release-Recapture experiment in the monitored area to compare and calibrate the model estimate produced by the N-mixture model approach. However, it is important to note that low and heterogenic capture rates among traps may represent a limiting factor for the exploitation of N-mixture model for mosquito surveillance (Veech, Ott, & Troy, 2016). In fact, the sticky traps used in the analysed case study were not highly effective in capturing a great fraction of adult mosquito population (Marini et al., 2010) and a great variation of capture rate was observed among traps, likely due to *Ae. albopictus*' great adaptability and plasticity in the oviposition behaviour (Hawley, 1988) and to the unrecorded presence of other factors affecting oviposition habitat selection (Fader & Juliano, 2014). Improved trap performance both in terms of capture rate and data collection would make N-mixture model estimates more reliable. Nevertheless, these variations in capture/detection process could be explicitly modelled in a Beta-Binomial process within the N-mixture approach. Beta-Binomial capture/detection process results showed that *Ae. albopictus* is mainly captured in traps positioned near small vegetated area, consistent with the species' preference for green spot as resting and oviposition sites (Bartlett-Healy et al., 2012; Crepeau et al., 2013; Manica et al., 2016). The amount of water in the trap also resulted associated with the capture/detection

probability as oviposition and consequently the probability of approaching the trap is expected to be lower when the trap water level is low (Unlu, Farajollahi, Strickman, & Fonseca, 2013).

In conclusion, this study represents one of the first applications of the N-mixture approach to species of epidemiological relevance. Public Health authorities would benefit from the introduction of novel statistical method for the estimation of mosquito population absolute abundance. Shifting the perspective from methods that estimate relative vector abundance to methods that focus on absolute abundance could help fill the gap between simple counts of mosquito collected in a monitoring perspective and inference capability of the actual biting population, in order to assess the actual risk of pathogen transmission, as well as the effectiveness of control interventions.

AUTHORS' CONTRIBUTIONS

M.M., A.S.o., F.F., B.C., A.d.T. and M.B. conceived the ideas and designed methodology; B.C., F.F., A.S.o. and A.S.c. collected the data; M.M., F.F. and M.B. analysed and interpreted data; M.M., A.d.T., M.B. and R.R. wrote the manuscript; B.C. and A.S.o. critically revised the manuscript. All authors gave final approval for publication.

DATA AVAILABILITY STATEMENT

Data available via the Figshare Repository <https://doi.org/10.6084/m9.figshare.8118035.v1> (Manica et al., 2019). Jags code available from the Figshare Repository. To access the project, go to https://figshare.com/projects/Applying_the_N-mixture_model_approach_to_estimate_mosquito_population_abundance_from_monitoring_data/61355.

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SUPPORTING INFORMATION

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